Operating System

4th Semester, Academic Year 2025

Date:23-04-2025

Name: Rithvik Rajesh Matta	SRN: PES2UG23CS485	Section H

4th Unit Banana Problem

Q3. Write a C Program to demonstrate file operations using system calls –open the file in write mode, write content up to 20 characters, and perform SEEK_END and SEEK_SET operations with the same file

code:

```
#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

#include <string.h>


int main() {

    int fd;

    char buffer[] = "Hello from system call!";
```

```
    char filePath[] = "demo.txt";

    fd = open(filePath, O_WRONLY | O_CREAT | O_TRUNC,
0644);

    if (fd < 0) {
        perror("Error opening file");
        return 1;
    }

    if (write(fd, buffer, 20) < 0) {
        perror("Error writing to file");
        close(fd);
        return 1;
    }

    off_t endPos = lseek(fd, 0, SEEK_END);
    if (endPos == -1) {
        perror("Error with SEEK_END");
        close(fd);
```

```c
        return 1;
    }
    printf("File pointer moved to end: %ld\n", (long)endPos);

    off_t startPos = lseek(fd, 0, SEEK_SET);
    if (startPos == -1) {
        perror("Error with SEEK_SET");
        close(fd);
        return 1;
    }
    printf("File pointer moved to start: %ld\n", (long)startPos);

    close(fd);
    return 0;
}
```

screenshot:

```
rithvikmatta@penguin:~/sem4/OS/banan prob 4$ gcc code.c
rithvikmatta@penguin:~/sem4/OS/banan prob 4$ ./a.out
File pointer moved to end: 20
File pointer moved to start: 0
```

If found plagiarized, I will abide with the disciplinary action of the University.

Signature:Rithvik Matta

Name: Rithvik Rajesh Matta

SRN:PES2UG23CS485

Section: H

Date:23/04/2025